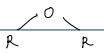
Lecture 17

Thursday, 30 March 2023 10:30 AM

Term Projects:

Epoxidation: $-C=C- \rightarrow$ (Epoxide) $-C(-O-)C-$

Ether:



General:

1) Literature survey: key papers

2) Calculations: mostly structural & thermodynamics but basic transition state for exp. students

HER on Pt (111) surface; + No water

1) $H^+ + e^- + * \leftrightarrow H^*$ (Don't need the kinetics but just the thermodynamics)

2) $2H^* \longleftrightarrow H_2(g) + 2*$ (Long-Hinshelwood step)

BE_H vs. θ_H (coverage) (Binding energy of H) $\Rightarrow$ Calculate H_2 formation barrier vs θ_H

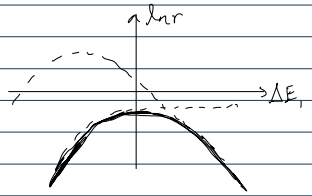
+ Surf-graph algorithm (Help w/ symmetry)

Then make a micro-kinetic model !!

Scaling Relationships & Volcano plots

1) $A_2(g) + 2* \longleftrightarrow 2A^*$

2) $A^* + B(g) \longleftrightarrow AB(g) + *$



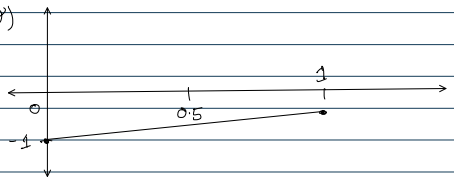
Interactions b/w adsorbates:

+ Surface phase diagram

+ Isotherms

+ Difficult to find relationships that are general enough to describe all of a volcano plot

+ Coverage-dependent binding energies (if deviations from mean field are small):



$$\Delta E_1 = \frac{E(N) - E(N-1)}{1} - E_{A(g)}$$

differential energy (given that there are already some adsorbates on the surface)

More weakly adsorbed @ higher coverages (more endothermic due to repulsion)

ex: $\Delta E_1(\theta) = \Delta E_1(\theta = 0) + \underbrace{0.75\theta}_{\text{Arbitrary slope}}$

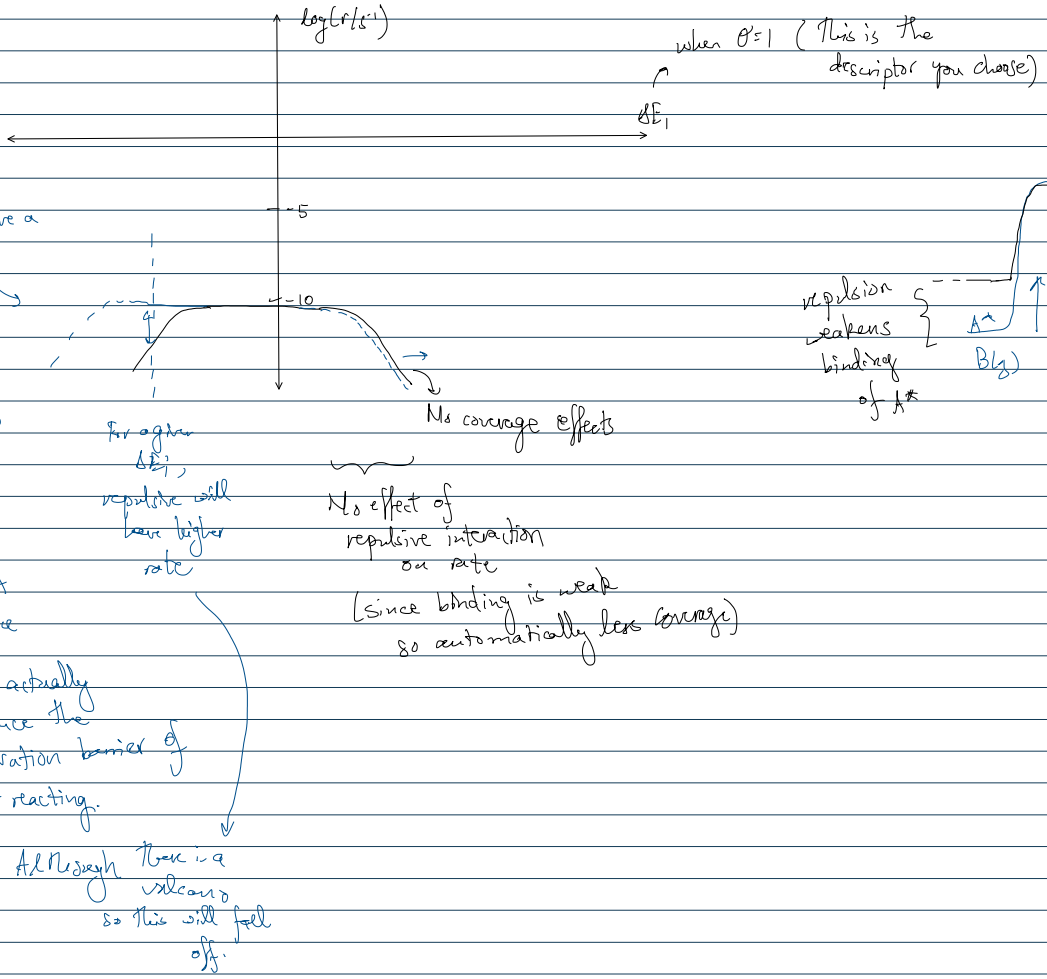
If we assume that BEP does not change w/ coverage (slope & intercept are same):

$$E_{a_1}(\theta) = \alpha_1 \Delta E_1(\theta) + \beta_1$$

If we assume that BEP does not change w/ coverage (slope & intercept are same):

$$E_{a_i}(\theta) = \alpha_i \Delta E_i(\theta) + \beta_i$$

Do not depend on coverage (although not always true)



Usually consider interactions b/w MASI & other intermediates on surface

Ammonia Synthesis:

1) $N_2(g) + 2* \leftrightarrow 2N^*$
2) $N^* + H^* \leftrightarrow NH^* + *$
3) $NH^* + H^* \leftrightarrow NH_2^* + *$
4) $NH_2^* + H^* \leftrightarrow NH_3(g) + 2*$
5) $H_2(g) + 2* \leftrightarrow 2H^*$

2 competing RLS

Limiting Analysis

overall reaction

1) 1 is RLS:

$$r = 2k_1 P_{N_2}(1-\theta)$$

Kinetics Analysis

1) 1 is RLS:
(from earlier analysis)

$$\Gamma_1 = \frac{2k_1 P_{H_2}(1-\theta)}{\left[1+\sqrt{K_5 P'_{H_2}}+\frac{P_{NH_3}}{K_4\sqrt{K_5 P'_{H_2}}}+\frac{P'_{NH_3}}{K_3K_4K_5P'_{H_2}}+\frac{P'_{NH_3}}{K_3K_4K_5(K_5P'_{H_2})^{1/2}}\right]^2}$$

2) Step 3 is RLS:

$$\Gamma_3 = \frac{k_3K_2K_1^{1/2}K_5\,P'^{1/2}_{N_2}P'_{H_2}(1-\sqrt{\theta})}{\left[1+\sqrt{K_5P'_{H_2}}+\sqrt{K_1P'_{N_2}}+K_2\sqrt{K_1P'_{H_2}}\sqrt{K_5P'_{H_2}}+\frac{P'_{NH_3}}{K_4\sqrt{K_5P'_{H_2}}}\right]^2}$$

Parameters: $P'_{H_2}=75$, $P'_{N_2}=25$

$S^\circ_{H_2} \approx 192$ J/mol K ; $S^a_{NH_3(g)} \approx 193$ J/mol K
(From tables or partition function approaches)

↳ (Ignore S of surface species): $\Delta S^\circ_1 \approx -192$, $\Delta S^\circ_2 \approx \Delta S^\circ_3 \approx 0$, $\Delta S^\circ_4 = 193$, $\Delta S^\circ_5 \approx -131$

$E_{a_1} \approx 0.9\,\Delta E_1 + 1.34$ eV (RPBE)
↳ The defects/steps are taken as active sites

↳ 4 adsorption intermediates
↳ Can we express all energies in terms of ΔE_1?

Many parameter are correlated:

$\Delta E_{NH_2} = 0.37\,\Delta E_N - 1.26$
$\Delta E_{NH} \approx 0.67\,\Delta E_N - 1.04$
} DFT RPBE

Note: $\Delta E_{NH^*} \equiv \Delta E_{NH^*} - E_* - E_{NH_3(g)}$ ← Definition

These relations result from bond order conservation: Thus, $K_2 = e^{\frac{-\Delta E_2 - T\Delta S^\circ_2}{k_B T}}$ (need to find ΔE_2 expression)

↓

ΔE for reaction 2

$\Delta E_2 \equiv E_{NH^*} + E_* - E_{H^*} - E_{H^*}$ } expression in terms of ΔE_1